

Evolutionary level

18
16
14
12
10
8
6
4
2
0

A178T

A47S

A52T

A60T

A74G

C99R

F104Y

H173R

L139F

L24I

N3K

Q144H

Q177H

Q54T

S102G

S142P

S143E

S181R

T87K

U49C

Y48F

Mutation

$\Delta\Delta G^\ddagger$ (kcal mol⁻¹)

10

5

0

-5

-10